Technical Data Sheet

(TDS)

TINMORRY Metallic PLA

Version: V1.0

• Introduction

TINMORRY Metallic PLA is a PLA filament designed to deliver a realistic metallic appearance. The filament and printed parts have a smooth, refined surface, while metallic pearlescent fillers are evenly blended into the base material to produce a natural, warm metallic sheen. Changes in viewing angle create layered light reflections, enabling a convincing metal-like visual effect.

It retains PLA's easy-printing characteristics, with a low printing barrier and stable print formation, and is less prone to warping, stringing, or nozzle clogging. It is compatible with most desktop FDM 3D printers and is available in a range of metallic colors at an accessible price. Metallic PLA is suitable for decorative objects, creative models, display samples, and other applications requiring a metal-like appearance at lower cost.

• Product Specifications

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Weight	1000 g
Length	338 ± 5 m
Spool Material	PP plastic (heat resistant to approx. 100 °C)
Spool Dimensions	Outer diameter: 200 mm; hub opening diameter: 55 mm; Width: 62 mm

• Recommended Printing Parameters

Pre-print Drying Conditions	50-60 °C, 6-8 h (forced-air oven or filament dryer)
Printing Environment	≤ 35 °C, ≤ 20% RH (keep sealed in a container with desiccant for moisture protection)
Storage Conditions	≤ 30 °C, ≤ 20% RH (keep sealed with desiccant for moisture protection)
Compatible Support Material	Same material or dedicated PLA support material
Compatible Printer Type	Open-frame or enclosed
Compatible Nozzle Material	Brass, stainless steel, hardened steel, and other common nozzle materials
Compatible Nozzle Diameter	0.2, 0.4, 0.6, 0.8 mm
Compatible Build Plate	Textured PEI build plate, etc.
Build Plate Surface Preparation	Applying glue or adhesive spray is recommended
Recommended Nozzle Temperature	195-230 °C
Heated Bed Temperature	65-75 °C (depending on printer and build plate type)
Maximum Printing Speed	Approx. 270 mm/s (220 °C, Bambu Lab X1C, P1S, A1)
Chamber Temperature	< 40 °C

• Properties

The commonly used physical, mechanical, chemical, and printing properties of TINMORRY Metallic PLA have been tested and are listed in the tables below.

• Physical Properties

Item	Test Method	Data
Density	ρ = m / v	1.23 g/cm³
Saturated Moisture Absorption	25 °C, 55% RH, ambient air	0.42%
Melt Flow Index	210 °C, 2.16 kg	18.2 ± 2.3 g/10 min
Glass Transition Temperature	DSC, 10 °C/min	53 °C
Crystallization Temperature	DSC, 10 °C/min	N/A
Melting Temperature	DSC, 10 °C/min	155 °C
Vicat Softening Temperature	ISO 306, GB/T 1633	58 °C

Heat Deflection Temperature 1	ISO 75, 1.82 MPa	51 °C
Heat Deflection Temperature 2	ISO 75, 0.45 MPa	56 °C

• Mechanical Properties

Item	Test Method	Data
Tensile Strength XY	ISO 527, GB/T 1040	37 ± 3 MPa
Tensile Strength Z	ISO 527, GB/T 1040	31 ± 3 MPa
Young's Modulus XY	ISO 527, GB/T 1040	2573 ± 182 MPa
Young's Modulus Z	ISO 527, GB/T 1040	2364 ± 117 MPa
Elongation at Break XY	ISO 527, GB/T 1040	6.1% ± 1.8%
Elongation at Break Z	ISO 527, GB/T 1040	2.5% ± 0.8%
Flexural Strength XY	ISO 178, GB/T 9341	95 ± 3 MPa
Flexural Strength Z	ISO 178, GB/T 9341	62 ± 2 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	3481 ± 90 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	3570 ± 80 MPa
Impact Strength XY	ISO 179, GB/T 1043	16 ± 2.5 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	9 ± 1.2 kJ/m²

• Other Physical and Chemical Properties

Item	Data
Main Composition	PLA
Skin Irritation	None
Water Resistance	Resistant
Oil and Grease Resistance	Resistant to most common oils and greases
Organic Solvent Resistance	Not resistant to certain organic solvents
Weak Acid Resistance	Resistant to short-term exposure
Strong Acid Resistance	Not resistant
Weak Alkali Resistance	Not resistant
Strong Alkali Resistance	Not resistant
Flammability	Flammable
Odor During Printing	Almost odorless
Odor During Combustion	Mildly irritating odor

• Test Specimen Information

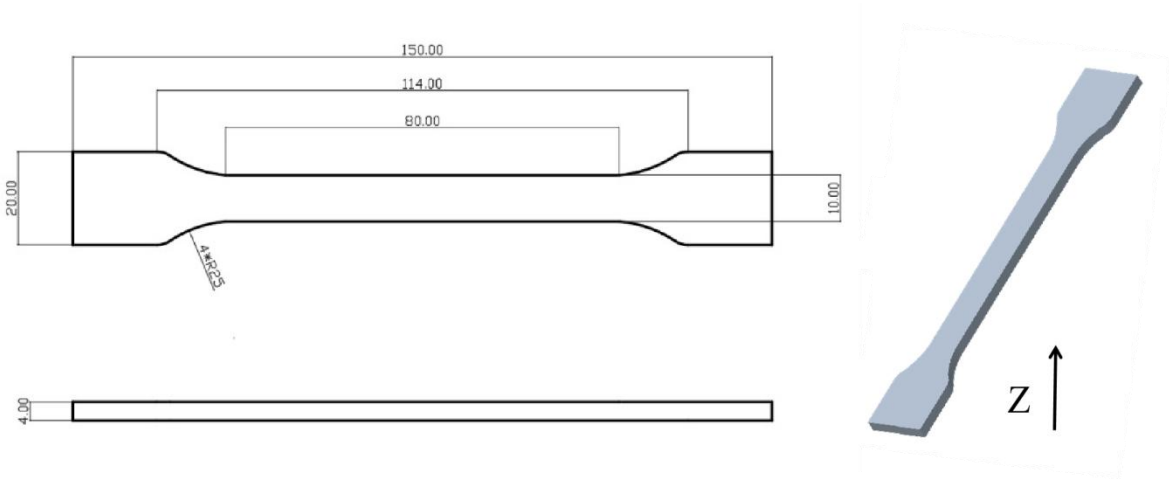
Mechanical Test Specimen Information	
Item	Data
Specimen Dimensions	See figure below
Nozzle Temperature	220 °C
Build Plate Temperature	65 °C

Printing Speed	XY: 258 mm/s; Z: 157 mm/s
Infill Density	100%
Infill Pattern	Concentric

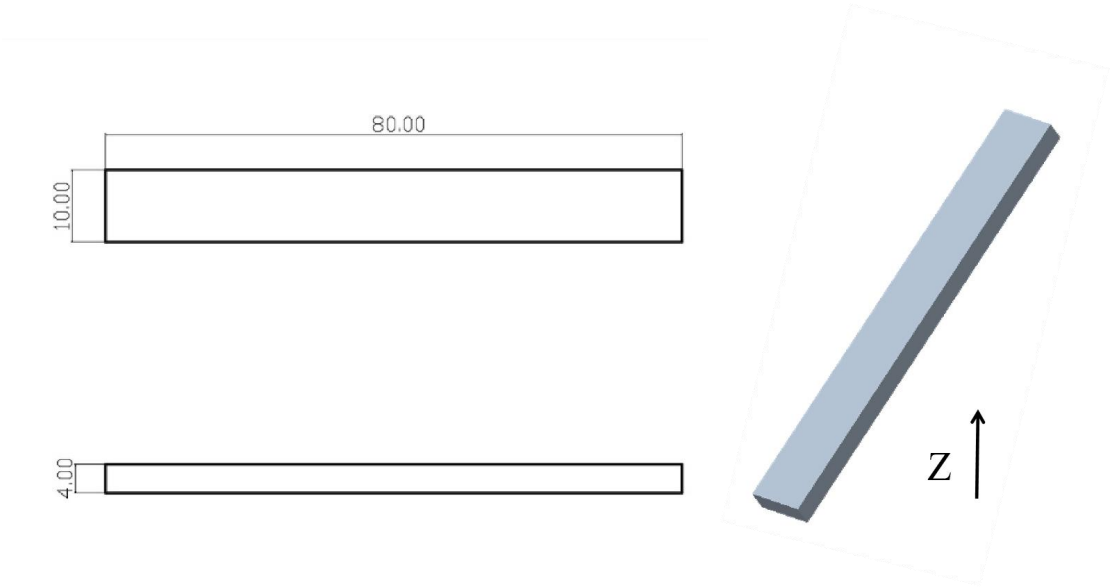
Notes:

1. The test specimens were printed using the key parameters listed above. Before testing, all specimens were conditioned indoors at approximately 25 °C for more than 12 h. Different ambient temperatures, fan speeds, and layer times during printing produce different cooling rates, which can affect the mechanical properties of the specimens, especially those printed in the Z direction. Different printers may also result in differences in printed-part properties.
2. For applications requiring higher print quality, thoroughly dry TINMORRY filament before printing and protect it from moisture during printing using a sealed container with desiccant (recommended relative humidity inside the container ≤ 25%) to reduce moisture-related printing issues. Refer to the "Recommended Printing Parameters" table for the recommended drying conditions. During drying, keep the filament and spool away from direct heat sources to avoid overheating damage. Do not use a kitchen oven or microwave oven.
2. Printed parts made from TINMORRY Metallic PLA may gain a small improvement in strength, stiffness, and heat resistance after annealing, but toughness may decrease slightly, and some parts may shrink or deform during annealing. The annealing result is also affected by part size, geometry, wall thickness, infill density, and other factors. If annealing is required, use a forced-air oven with uniform temperature. It is recommended to anneal at approximately 50 °C for about 4 h, then slowly increase the temperature to approximately 55 °C and hold for about 4-6 h. If obvious deformation occurs during the process, stop annealing immediately and use a lower annealing temperature for subsequent attempts.

1. Tensile Test



2. Flexural Test and Unnotched Impact Test



• Disclaimer

The data above were obtained by TINMORRY using specific printers, test methods, industry-standard specimens and methods, and specific equipment, and are provided for reference and comparison only. If the data change, TINMORRY will update this TDS, but may not be able to provide individual notice to users. In addition, the performance of 3D-printed parts depends not only on the intrinsic properties of the filament, but also on many other factors, including filament moisture content, printer type and condition, ambient temperature and ventilation, part geometry, and printing parameters. Users are responsible for the safety, legality, regulatory compliance, and performance of parts produced using TINMORRY 3D printing filament. TINMORRY is responsible for the quality of the filament sold, but assumes no responsibility for any damage or injury arising during its use or for the suitability of the filament for any particular application.